Experiment – 8: Laplace Transform, Inverse Transform, ROC and Stability

Objective 2:

Determination of Inverse Laplace Transform for Signal Reconstruction

Experimental Procedure:

- Define X(s) symbolically.
- Apply ilaplace to obtain x(t).
- Evaluate x(t) over 0≤t≤5 s.
- Plot the reconstructed time-domain signal. Observe plot
- Compare the plotted response with the analytical expression.

Code:

```
clc;
clear;
close all;

% Inverse Laplace Transform
% Requires Symbolic Math Toolbox

syms s t

% Laplace Transform
X = 1/((s+2)*(s+3));

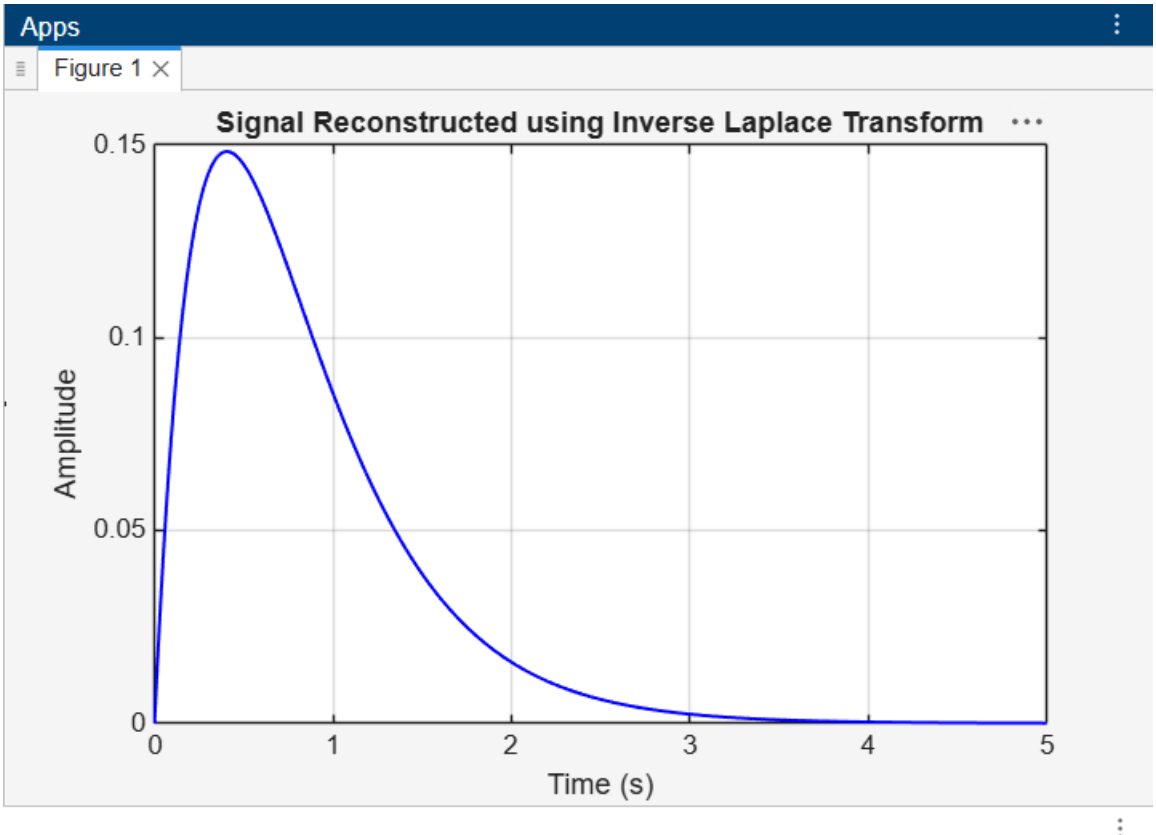
% Inverse Laplace Transform
x = ilaplace(X,s,t);

disp('Inverse Laplace Transform x(t) =');
pretty(x);
```

```
% Signal reconstruction
tt = 0:0.001:5;
xx = double(subs(x,t,tt));

% Plotting
figure;
plot(tt,xx,'b','LineWidth',1.2);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Signal Reconstructed using Inverse Laplace Transform');
```

output:



Objective 3:

To determine the Region of Convergence (ROC) of Laplace transforms and analyze its significance in characterizing continuous-time signals and systems. Expected Outcomes

Experimental Procedure:

• Define the pole locations.

• Plot the poles in the s-plane.

• Mark the vertical lines through the poles. Record result

• State the candidate ROC for each signal-sidedness case.

• Interpret the relationship between poles and ROC.

Code:

```
clc;
clear;
close all;

% Analysis of Region of Convergence (ROC)

% Poles of X(s) = 1/((s+2)(s+4))
p = [-2 -4];

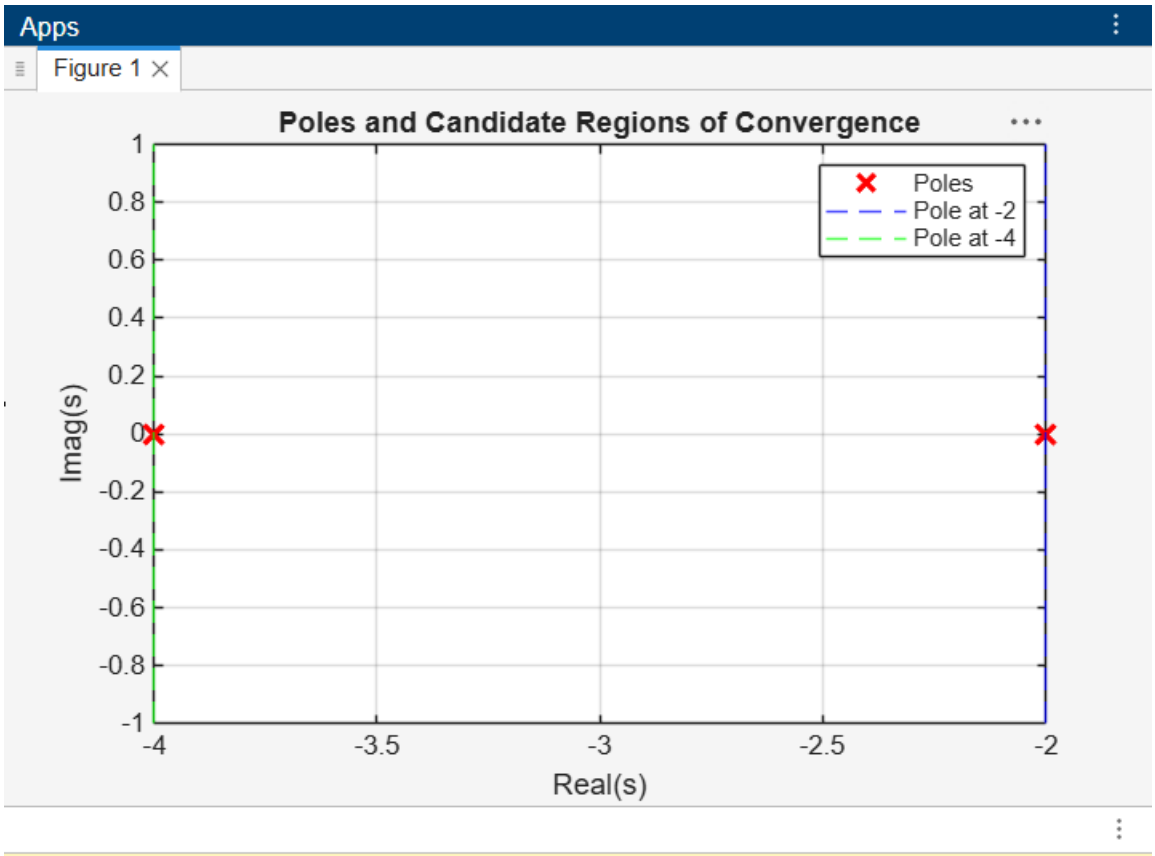
% Plot poles
figure;
plot(real(p),imag(p),'rx','MarkerSize',10,'LineWidth',2);
hold on;

xline(-2,'--b');
xline(-4,'--g');
```

```
grid on;
xlabel('Real(s)');
ylabel('Imag(s)');
title('Poles and Candidate Regions of Convergence');
legend('Poles','Pole at -2','Pole at -4');

% Display ROC
fprintf('Poles are at s = -2 and s = -4.\n');
fprintf('Right-sided ROC: Re(s) > -2.\n');
fprintf('Left-sided ROC: Re(s) < -4.\n');
fprintf('Two-sided ROC: -4 < Re(s) < -2.\n');
```

output:



RESULT:

Thus, the inverse Laplace Transform was successfully calculated, and the time-domain signal was reconstructed. The poles and possible Regions of Convergence (ROC) were also identified and plotted, and their significance in system analysis was studied.

Objective	**Assessment Criterion**	**Excellent – 4**	**Good – 3**	**Satisfactory – 2**	**Needs Improvement – 1**	**Weightage**	**Marks given**	**Marks obtained**
(i) Determine the inverse Laplace transform of a given frequency-domain representation and reconstruct the corresponding time-domain signal.	**Inverse Laplace Transform Evaluation**	Correctly expands X(s) using partial fractions (distinct, repeated and complex poles) and applies the standard transform pairs and properties. Obtains the exact time-domain expression x(t), including the unit step u(t) and any shifts, with correct constants.	Applies partial fractions and transform pairs correctly, with minor errors in residues, constants or step-function terms.	Performs basic inverse transforms but has noticeable errors in partial fractions or transform pairs, and needs guidance.	Unable to determine the inverse Laplace transform.	**20**	**2**	

	Time-Domain Reconstruction & MATLAB Implementation	Uses suitable MATLAB commands (residue, ilaplace, syms, impulse or step) with a proper time vector. Reconstructs and plots x(t) with titles, axis labels and grid, and verifies the numerical result against the analytical expression.	Reconstructs and plots the signal correctly, with minor errors in code, time range, parameters or labelling.	Generates a basic reconstruction with noticeable errors in code or plots, and needs assistance.	Unable to reconstruct the time-domain signal or produce meaningful plots.	**20**	**2**	
(ii) Determine the Region of Convergence (ROC) of Laplace transforms and analyze its significance in characterizing continuous-time signals	**ROC Determination**	Correctly determines the ROC for right-sided, left-sided, two-sided and finite-duration signals from the pole locations. States the ROC as a proper inequality on Re(s) and shows it on the s-plane	Determines the ROC correctly for most signals, with minor errors in the boundary, signal type or s-plane plot.	Determines the ROC for basic cases only, with noticeable errors, and needs guidance.	Unable to determine the ROC of the given Laplace transforms.	**20**	**2**	

and systems.		with poles and zeros marked.						
	ROC Significance for Signals and Systems	Clearly relates the ROC to the signal type, causality and BIBO stability (ROC contains the jω-axis), and shows that the same X(s) with different ROCs gives different time signals. Supports the reasoning with mathematical conditions and pole-zero evidence.	Relates the ROC to causality and stability correctly, with minor errors or limited justification.	Gives a basic explanation of the ROC's significance. Justification is incomplete and needs guidance.	Unable to explain the significance of the ROC.	**20**	**2**	
Interpret the Laplace-domain results and document the experiment.	**Engineering Analysis, Interpretation & Record**	Provides technically accurate interpretation of the results (pole locations, ROC and signal	Maintains a proper record with generally accurate interpretation, with minor	Record and interpretation are basic and incomplete, with some technical	Incomplete record, limited or incorrect interpretation of the results.	**20**	**2**	

		behaviour). Maintains a complete, well-organized record with code, outputs, plots and conclusions.	omissions.	inaccuracies.				
					Total	**100**	**10**	